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Cellular Biology

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DENT1020 Dental Science I

This lecture…

By the end of this lecture you should be able to:

1. Appreciate that cells are the building blocks of all biological units
2. Identify the components of the cell on a diagram and describe the different cell components
3. Outline the cell cycle, describing the phases of mitosis that are involved

Suggested additional reading:

- *Fehrenbach MJ & Popowics T. Illustrated Dental Embryology, Histology, and Anatomy. 4th edn. Missouri, USA. Elsevier Saunders. (Chapter 7)*

Why do we need to learn about cellular biology?

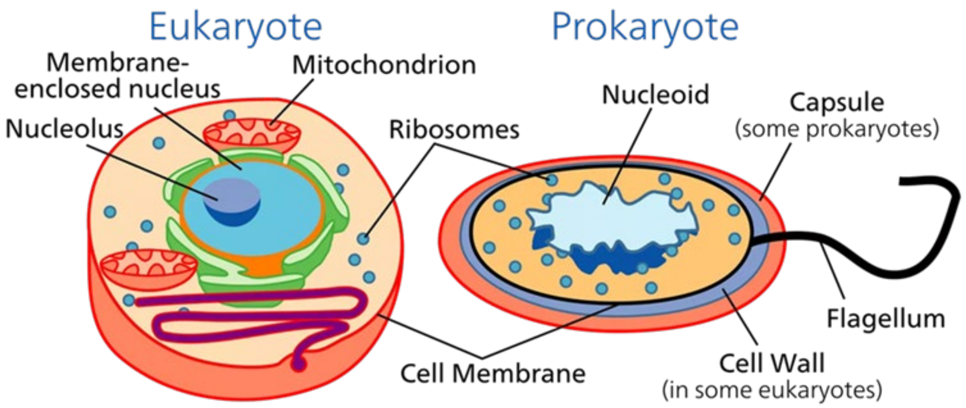
https://www.jisponline.com/articles/2010/14/3/images/JIndianSocPeriodontol_2010_14_3_195_75917_f1.jpg

Prokaryotic vs Eukaryotic Cells

- Cells are the basic structural and functional units of every organism
- They are of two distinct types: prokaryotic and eukaryotic
- All cells share certain basic features:
  - Bounded by a selective barrier (plasma membrane)
  - Contain a semifluid, jelly-like substance (cytosol) in which subcellular components are suspended
  - Contain chromosomes, which carry genes in the form of DNA
  - Have ribosomes, tiny complexes that make proteins according to instructions from the genes

Differences

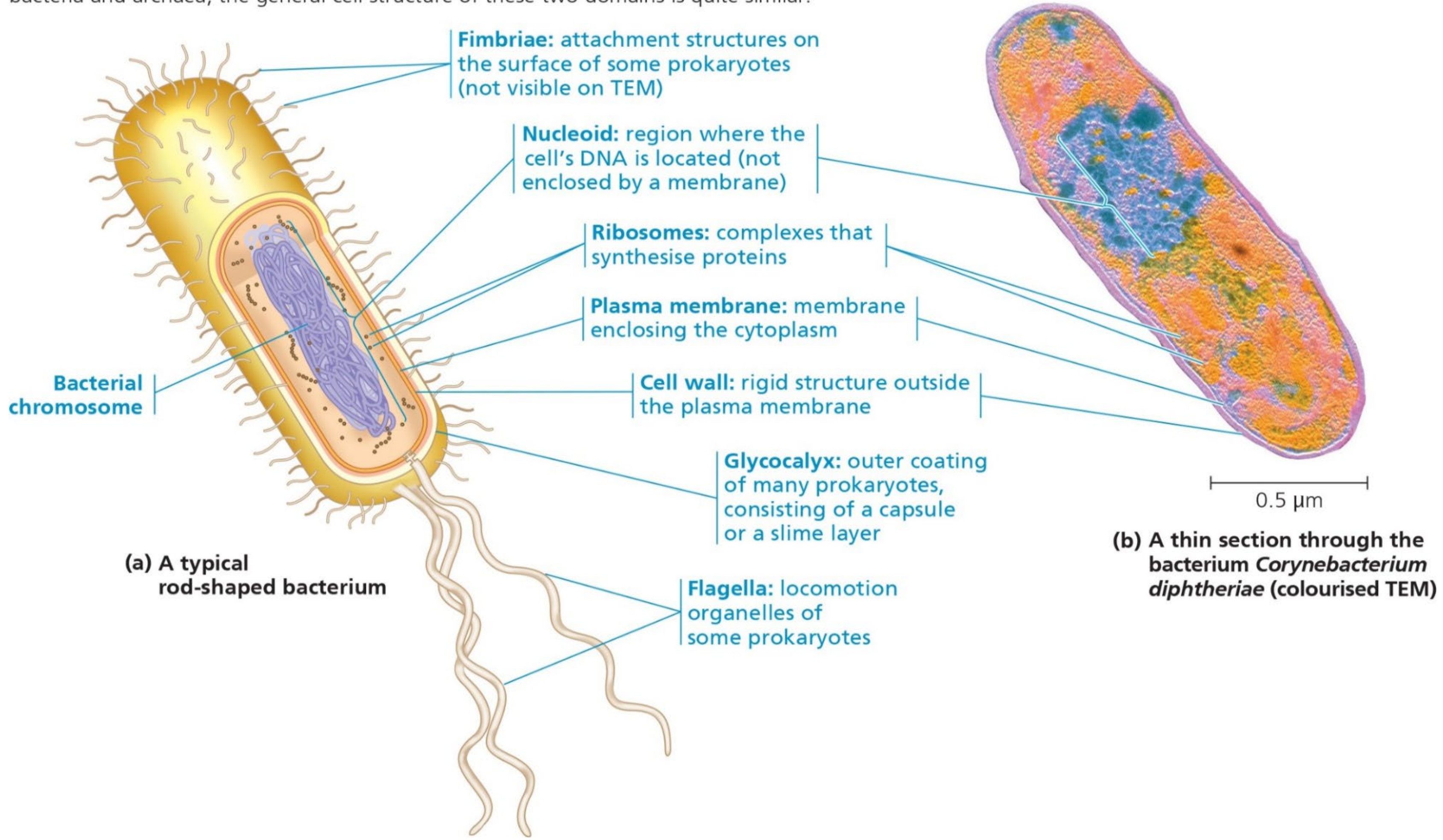
- A major difference between prokaryotes and eukaryotes is the location of their DNA
  - In eukaryotic cells most of the DNA is in the nucleus, which is bounded by a double membrane
  - In prokaryotic cells the DNA is concentrated in the nucleoid, a region that is not membrane-enclosed
- The presence of membrane-bound organelles
  - In eukaryotic cells a variety of organelles of specialised form and
    function are suspended in cytosol
  - These organelles are absent in almost all prokaryotic cells



https://www.thoughtco.com/endosymbiotic-theory-of-evolution-1224532

A Prokaryotic Cell

Figure 6.5 A prokaryotic cell. Lacking a true nucleus and the other membrane-enclosed organelles of the eukaryotic cell, the prokaryotic cell appears much simpler in internal structure. Prokaryotes include bacteria and archaea; the general cell structure of these two domains is quite similar.



(a) A typical rod-shaped bacterium

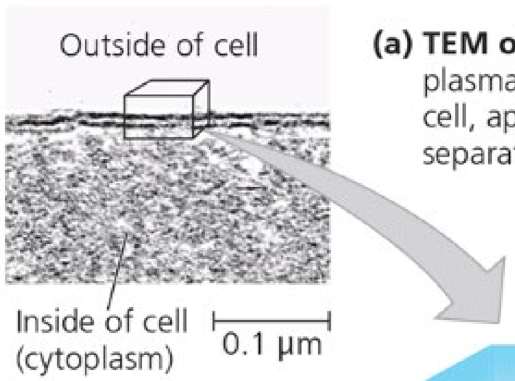
(b) A thin section through the bacterium *Corynebacterium diphtheriae* (colourised TEM)

From Urry LA et al. Campbell Biology, Australian and New Zealand edition. 11th edn. Sydney, Australia: Pearson Australia, 2017, p97

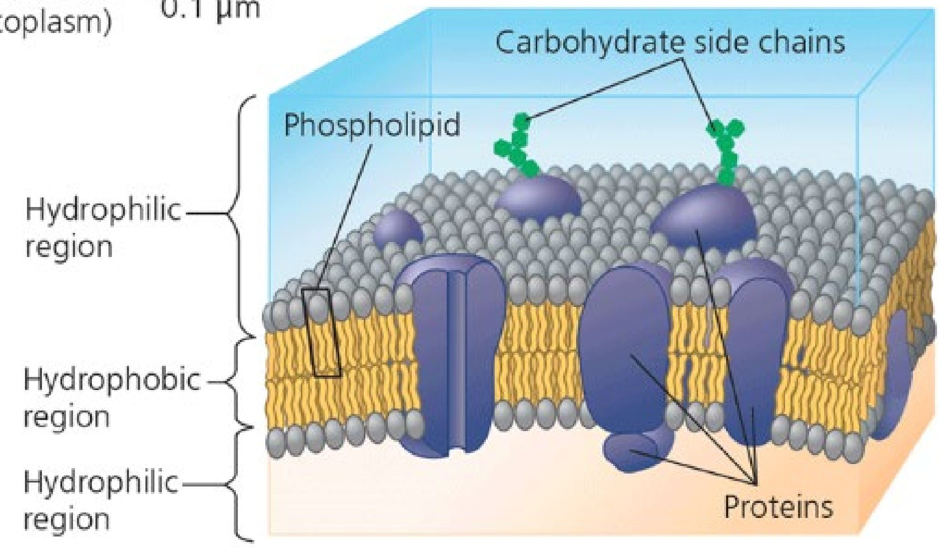
Cell Size

- Size is a general feature of cell structure that relates to function
- The logistics of carrying out cellular metabolism sets limits on cell size
- Eukaryotic cells are typically 10-100 µm in diameter.
- Metabolic requirements also impose theoretical upper limits on the size that is practical for a single cell
- The plasma membrane functions as a selective barrier that allows passage of enough oxygen, nutrients and waste to service the entire cell
- Larger organisms do not generally have larger cells than smaller organisms – they simply have more cells

Plasma Membrane



(a) TEM of a plasma membrane. The plasma membrane, here in a red blood cell, appears as a pair of dark bands separated by a light band.



(b) Structure of the plasma membrane

- Encloses the cells contents, and regulates the passage of molecules across it
- Composed of a phospholipid bilayer, each lipid has two hydrophobic fatty acid tails and a hydrophilic phosphate head
- Semipermeable (selective) membrane and will only allow certain ions or molecules to pass

From Urry LA et al. Campbell Biology, Australian and New Zealand edition. 11th edn. Sydney, Australia: Pearson Australia, 2017

Eukaryotic Cells

A eukaryotic cell has extensive, elaborately arranged internal membranes that divide the cells into **compartments**

These compartments provide different local environments that support specific metabolic functions, so incompatible processes can occur simultaneously in a single cell.

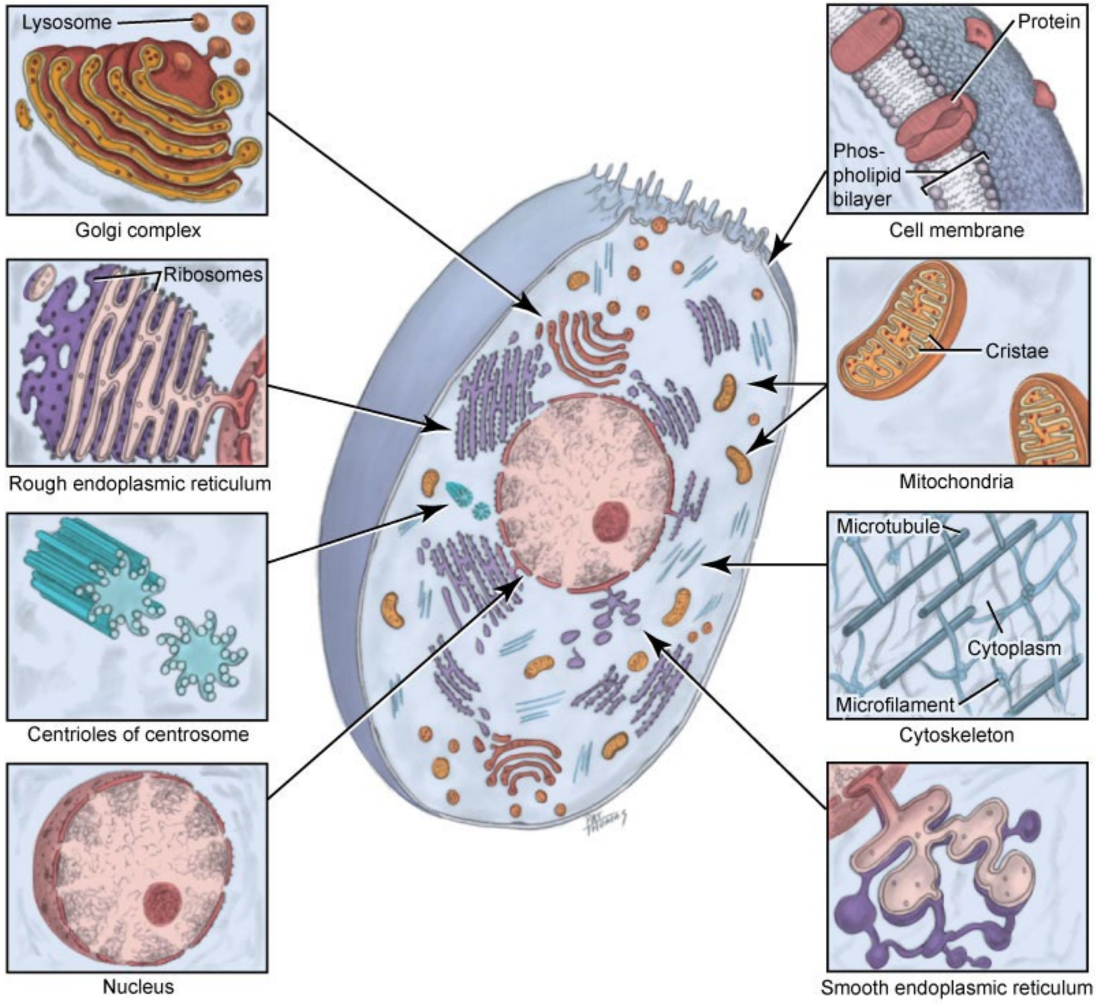


Fig. 7-2. The cell with its organelles and cell membrane examined.

From Fehrenbach MJ & Popowics T. Illustrated Dental Embryology, Histology, and Anatomy. 4th edn. Missouri, USA. Elsevier Saunders. 2016, p78

Take home activity on Blackboard
Bring to workshop

On the diagram of a cell with its cell membrane and organelles:

1. Identify and label the cell characteristics numbered 1-21

2. Describe the function of each characteristic

How does the internal organisation of eukaryotic cells allow them to perform the functions of life?

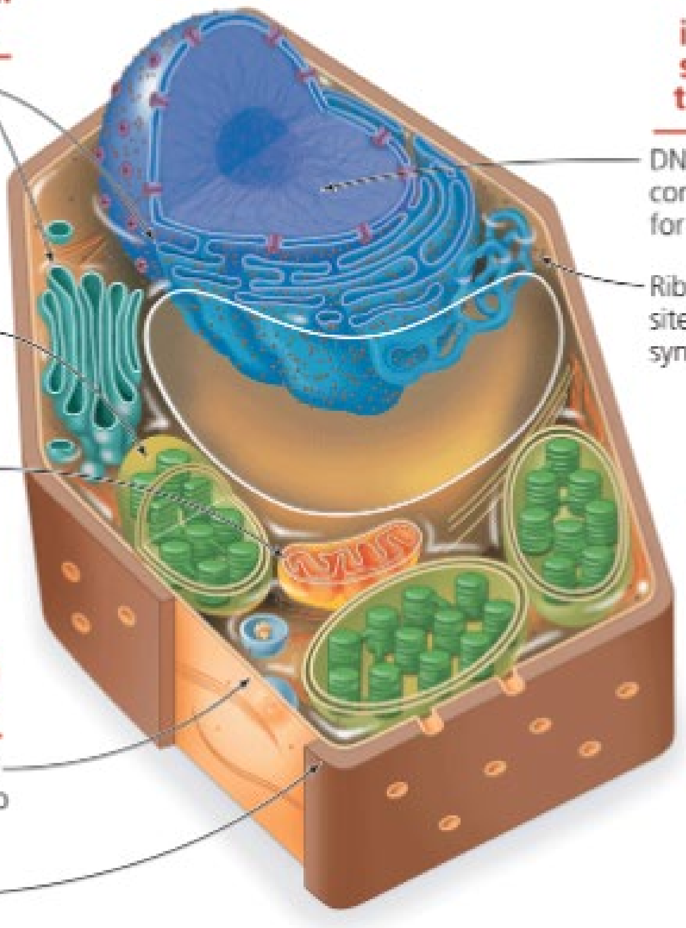
Internal membranes divide a cell, such as this plant cell, into compartments where specific chemical reactions occur.

Energy and matter transformations

A system of internal membranes synthesises and modifies proteins, lipids, and carbohydrates.

Chloroplasts convert light energy to chemical energy.

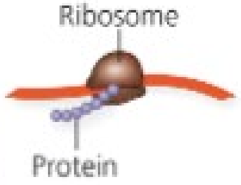
Mitochondria break down molecules, generating ATP.



Genetic information storage and transmission

DNA in the nucleus contains instructions for making proteins.

Ribosomes are the sites of protein synthesis.



Interactions with the environment

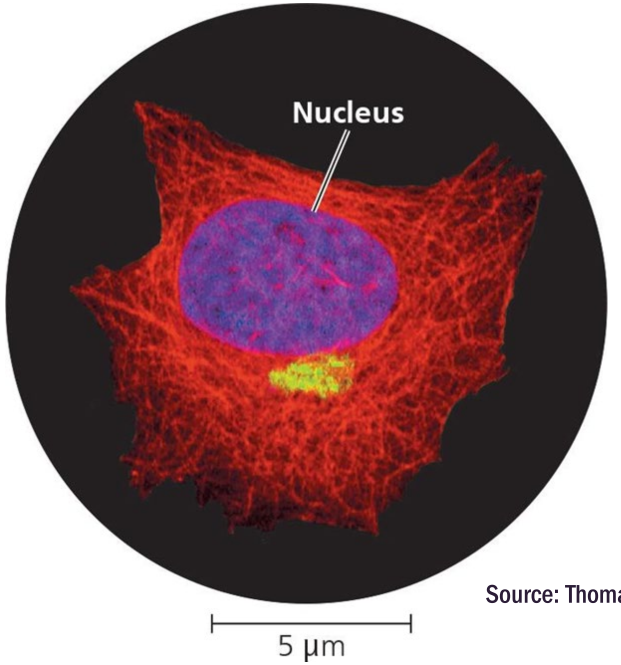
The plasma membrane controls what goes into and out of the cell.

Plant cells have a protective cell wall.

Campbell Biology (12th Australian and New Zealand edition.). Pearson Education Australia.2022

Cell nucleus

- Largest, densest and most conspicuous organelle in the cell (~5µm)
- Found in all cells in the body except mature red blood cells
- Stores the genetic code of the cell
- The nuclear envelope encloses the nucleus, separating its contents from the cytoplasm
- DNA is organised into discrete units called chromosomes.
- Single nucleus vs multinucleated



Source: Thomas Deerinck/Mark Ellisman/NCMIR

From Urry LA et al. Campbell Biology, Australian and New Zealand edition. 11th edn. Sydney, Australia: Pearson Australia, 2017, p102

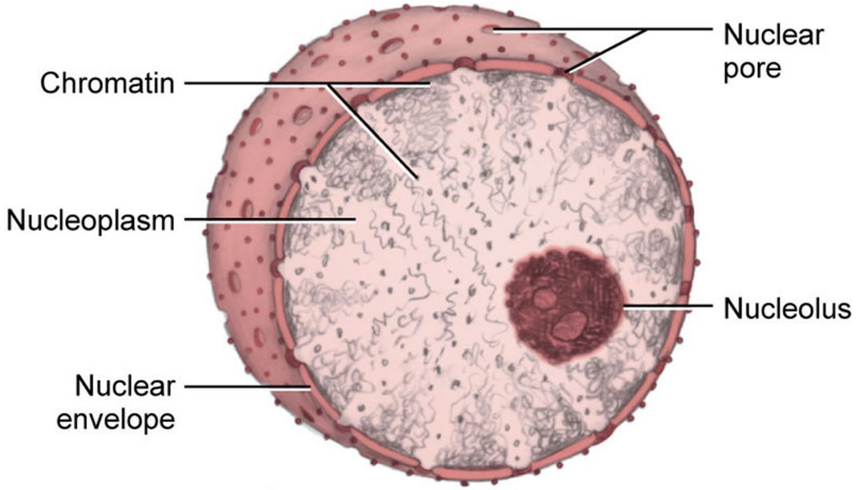


Fig. 7-3. Nucleus and its various components: chromatin, nucleoplasm, nuclear envelope, nuclear pore, and nucleolus.

From Fehrenbach MJ & Popowics T. Illustrated Dental Embryology, Histology, and Anatomy. 4th edn. Missouri, USA. Elsevier Saunders. 2016, p79

The key roles of cell division

"Where a cell exists, there must have been a pre-existing cell, just as the animal arises only from an animal and the plant only from a plant"

(Rudolf Virchow, 1855)

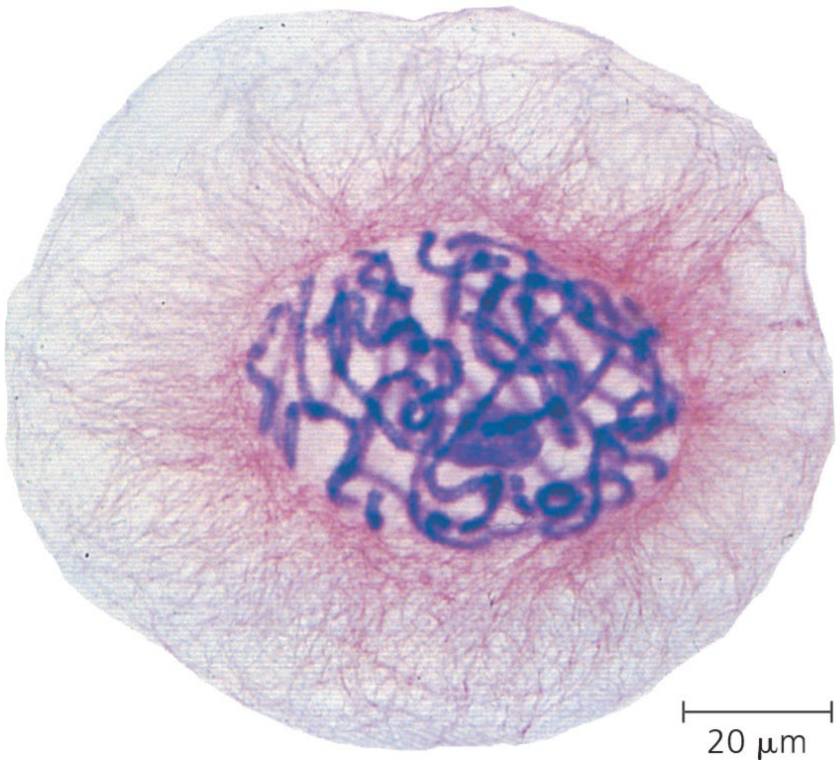
Cellular organisation of genetic material

Genome – the genetic material of an organism or a virus

Chromosomes – a cellular structure consisting of one DNA molecule and associated protein molecules.

Human somatic cells each contain 46 chromosomes (two sets of 23 – one inherited from each parent)

Figure 12.3 Eukaryotic chromosomes. Chromosomes (stained purple) are visible within the nucleus of this cell from an African blood lily. The thinner red threads in the surrounding cytoplasm are the cytoskeleton. The cell is preparing to divide (LM).



From Urry LA et al. Campbell Biology, Australian and New Zealand edition. 11th edn. Sydney, Australia: Pearson Australia, 2017, p237

The Cell Cycle

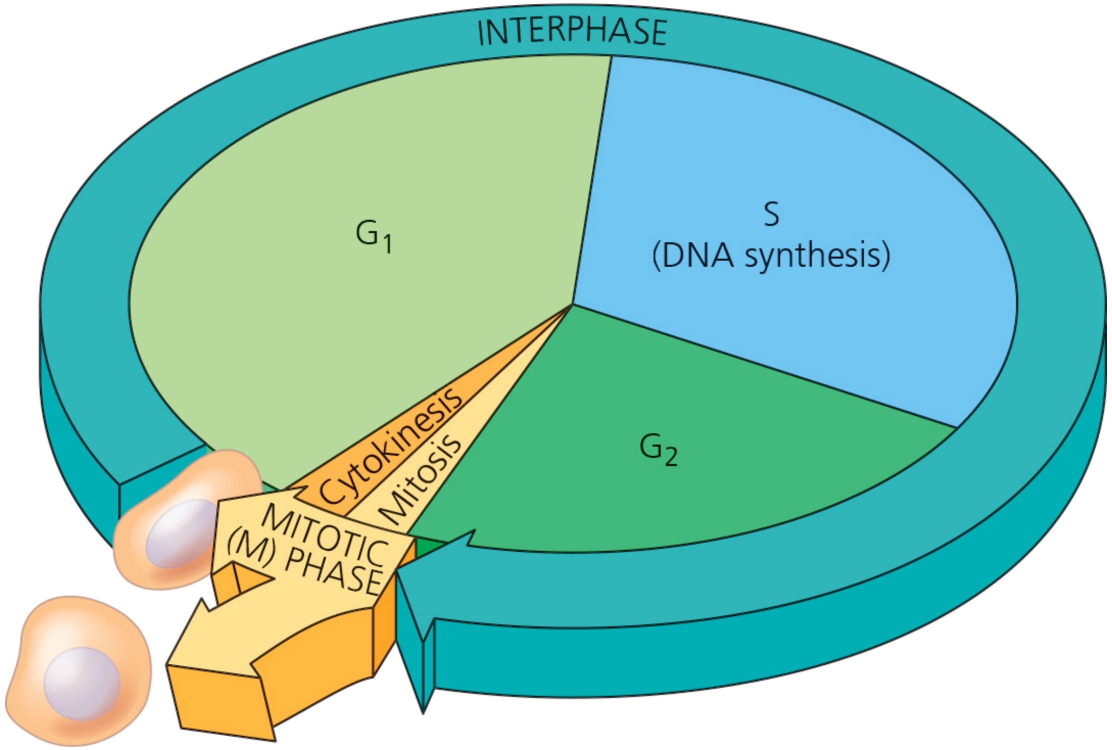
The cell cycle is an ordered sequence of events in the life of the cell, from its origin in the division of a parent cell until its own division into two.

Composed of:

Interphase (including G1, S and G2 phases)

Mitotic phase (M) (including mitosis and cytokinesis)

Figure 12.6 The cell cycle. In a dividing cell, the mitotic (M) phase alternates with interphase, a growth period. The first part of interphase (G_1) is followed by the S phase, when the chromosomes duplicate; G_2 is the last part of interphase. In the M phase, mitosis distributes the daughter chromosomes to daughter nuclei, and cytokinesis divides the cytoplasm, producing two daughter cells.



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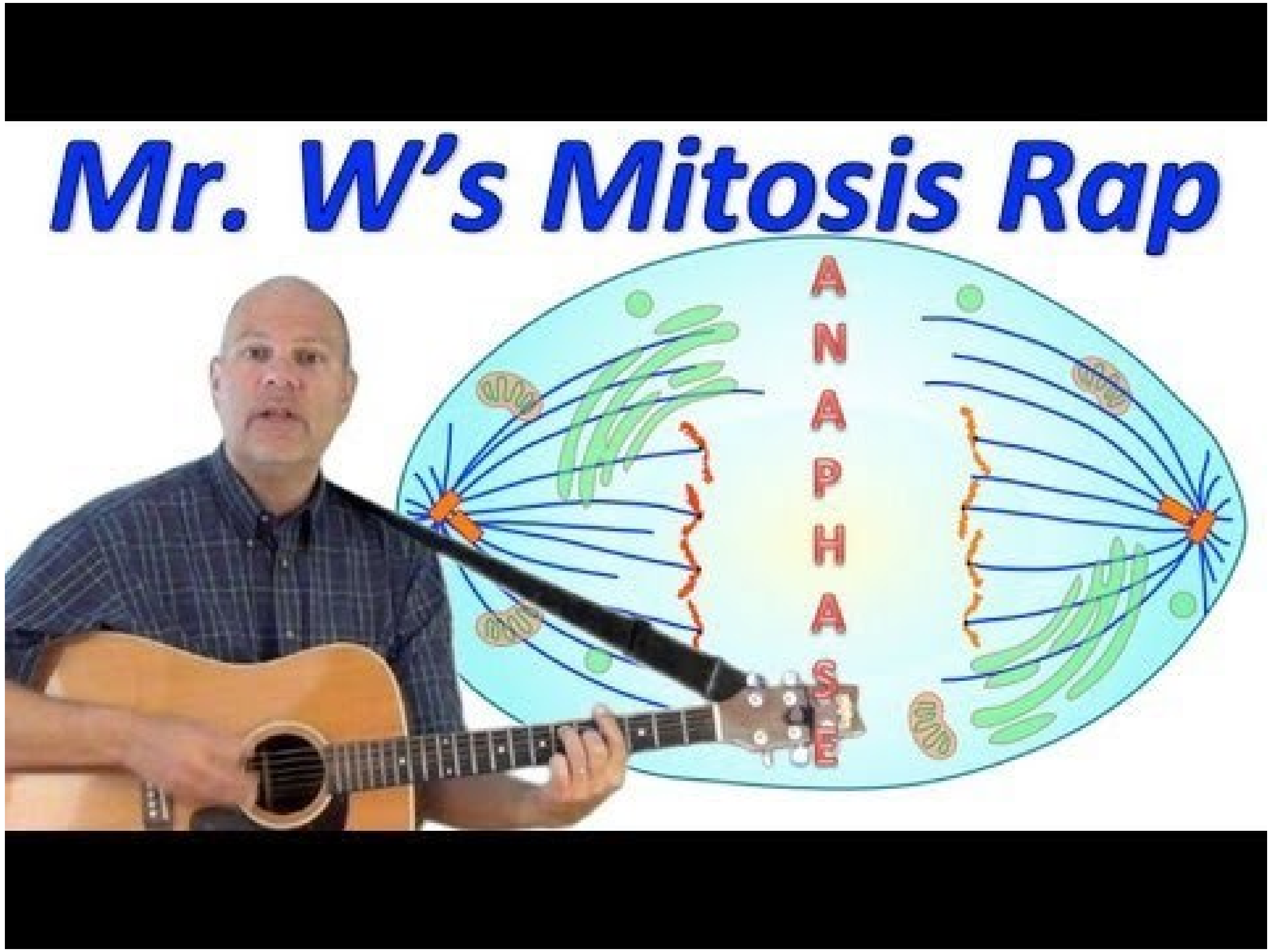
Mitosis

- **Mitosis is conventionally broken into five stages**

Mitosis

- Mitosis is conventionally broken into five stages



Mitosis in motion

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Something to think about:

Prophase: Coil to shape

Metaphase: Align and attach

Anaphase: Divide and drag

Telophase: Contract and expand

Cytokinesis: Separate and go your own way

Why is genetics relevant to dentistry?

- **Hereditary factors contribute to caries, periodontal disease, oral cancer, absent or malformed teeth, and other common oral disorders**
- **There are implications of systemic genetic disease on oral health care**
- **Genetic anomalies can influence growth and development**

- Knowing the molecular biology of bone, periodontal structures, salivary glands, and tooth development will lead to innovative treatment approaches that differ from the current surgical techniques used
- Tissue engineering is already making significant strides in cell manipulation and in developing tissue such as skin, bones, and cartilage.
- Advances in drug delivery, gene therapy and biopharmaceuticals will create additional new therapeutic methods that are vastly different from those currently used.

Inheritance of genes

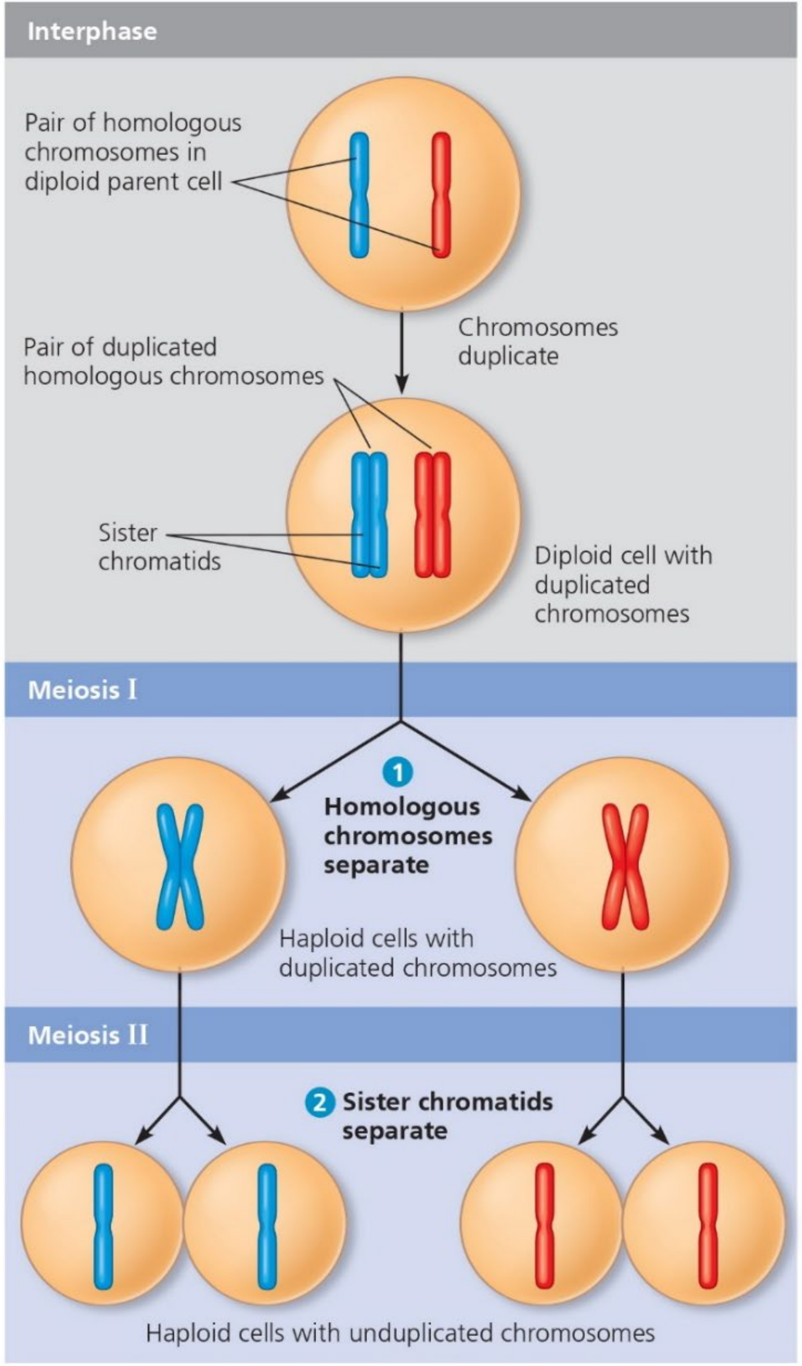
- Genetics is the scientific study of heredity and inherited variation.
- The hereditary units are called genes.
- The transmission of hereditary traits has its molecular basis in the replication of DNA, which produces copies of genes that can be passed from parents to offspring.
- In animals, reproductive cells called gametes are the vehicles that transmit genes from one generation to the next.
- Humans have 46 chromosomes in their somatic cells
- Each chromosome consists of a single long DNA molecule, elaborately coiled in association with various proteins
- A gene's specific location along the length of a chromosome is called the gene's locus

Chromosomes in human cells

- The 46 chromosomes in each somatic cell are actually 23 pairs
- The two chromosomes of a pair have the same length, centromere position, and staining pattern – they are called homologous chromosomes (or homologues)
- Both chromosomes of each pair carry genes controlling the same inherited characteristics
- The X and Y chromosomes are called sex chromosomes; the other chromosomes are called autosomes
- Human females have a homologous pair of X chromosomes (XX) while males have one X and one Y chromosome (XY)
- The occurrence of pairs of homologous chromosomes in each human somatic cell is a consequence of our sexual origins
- Any cell with two chromosome sets is called a diploid cell; gametes contain a single set of chromosomes and are called haploid cells

Meiosis

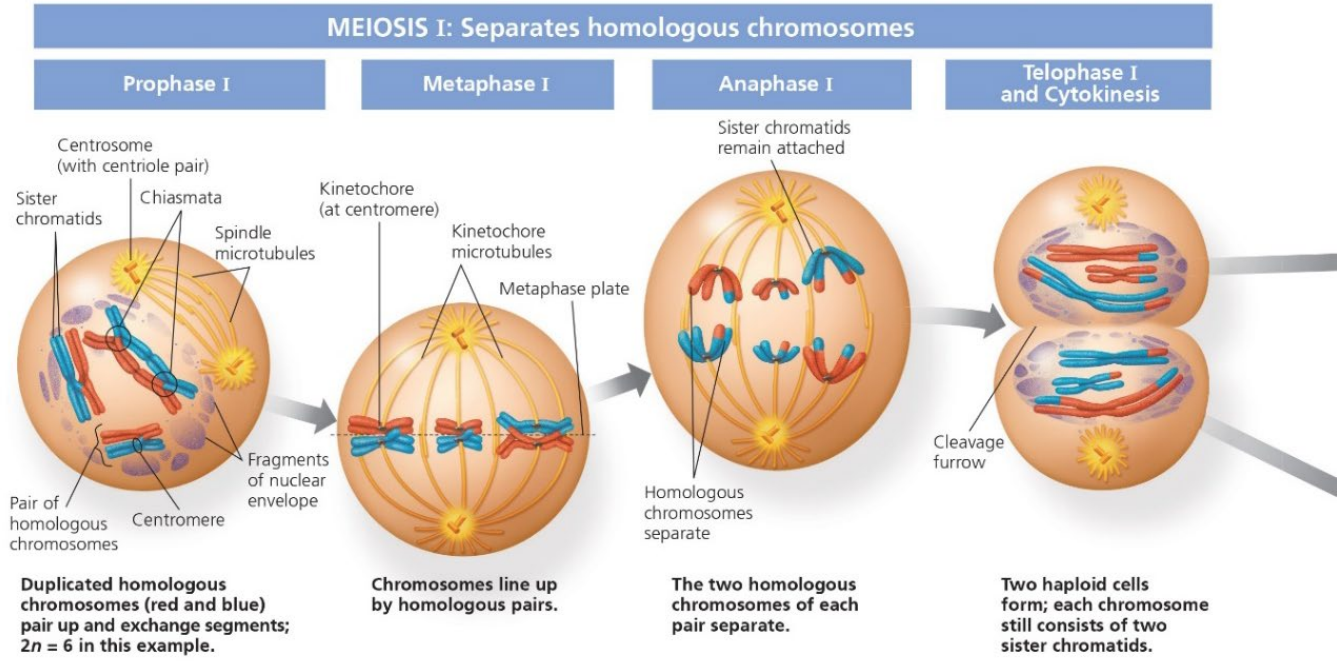
- Meiosis is preceded by the duplication of chromosomes
- This is followed by two consecutive cell divisions, meiosis I and meiosis II
- The two divisions result in four daughter cells, each with only half as many chromosomes as the parent cell



DRAW IT ➤ *Redraw the cells in this figure using a simple double helix to represent each DNA molecule.*

From Urry LA et al. Campbell Biology, Australian and New Zealand edition. 11th edn. Sydney, Australia: Pearson Australia, 2017, p261

Meiosis



Prophase I

- Centrosome movement, spindle formation, and nuclear envelope breakdown occur as in mitosis. Chromosomes condense progressively throughout prophase I.
- During early prophase I, before the stage shown above, each chromosome pairs with its homologue, aligned gene by gene, and **crossing over** occurs: The DNA molecules of nonsister chromatids are broken (by proteins) and are rejoined to each other.
- At the stage shown above, each homologous pair has one or more X-shaped regions called **chiasmata** (singular, *chiasma*), where crossovers have occurred.
- Later in prophase I, microtubules from one pole or the other attach to the kinetochores, one at the centromere of each homologue. (The two kinetochores on the sister chromatids of a homologue are linked together by proteins and act as a single kinetochore.) Microtubules move the homologous pairs towards the metaphase plate (see the metaphase I diagram).

Metaphase I

- Pairs of homologous chromosomes are now arranged at the metaphase plate, with one chromosome of each pair facing each pole.
- Both chromatids of one homologue are attached to kinetochore microtubules from one pole; the chromatids of the other homologue are attached to microtubules from the opposite pole.

Anaphase I

- Breakdown of proteins that are responsible for sister chromatid cohesion along chromatid arms allows homologues to separate.
- The homologues move towards opposite poles, guided by the spindle apparatus.
- Sister chromatid cohesion persists at the centromere, causing chromatids to move as a unit towards the same pole.

Telophase I and Cytokinesis

- When telophase I begins, each half of the cell has a complete haploid set of duplicated chromosomes. Each chromosome is composed of two sister chromatids; one or both chromatids include regions of nonsister chromatid DNA.
- Cytokinesis (division of the cytoplasm) usually occurs simultaneously with telophase I, forming two haploid daughter cells.
- In animal cells like these, a cleavage furrow forms. (In plant cells, a cell plate forms.)
- In some species, chromosomes decondense and nuclear envelopes form.
- No chromosome duplication occurs between meiosis I and meiosis II.

From Urry LA et al. Campbell Biology, Australian and New Zealand edition. 11th edn. Sydney, Australia: Pearson Australia, 2017, p263

Meiosis

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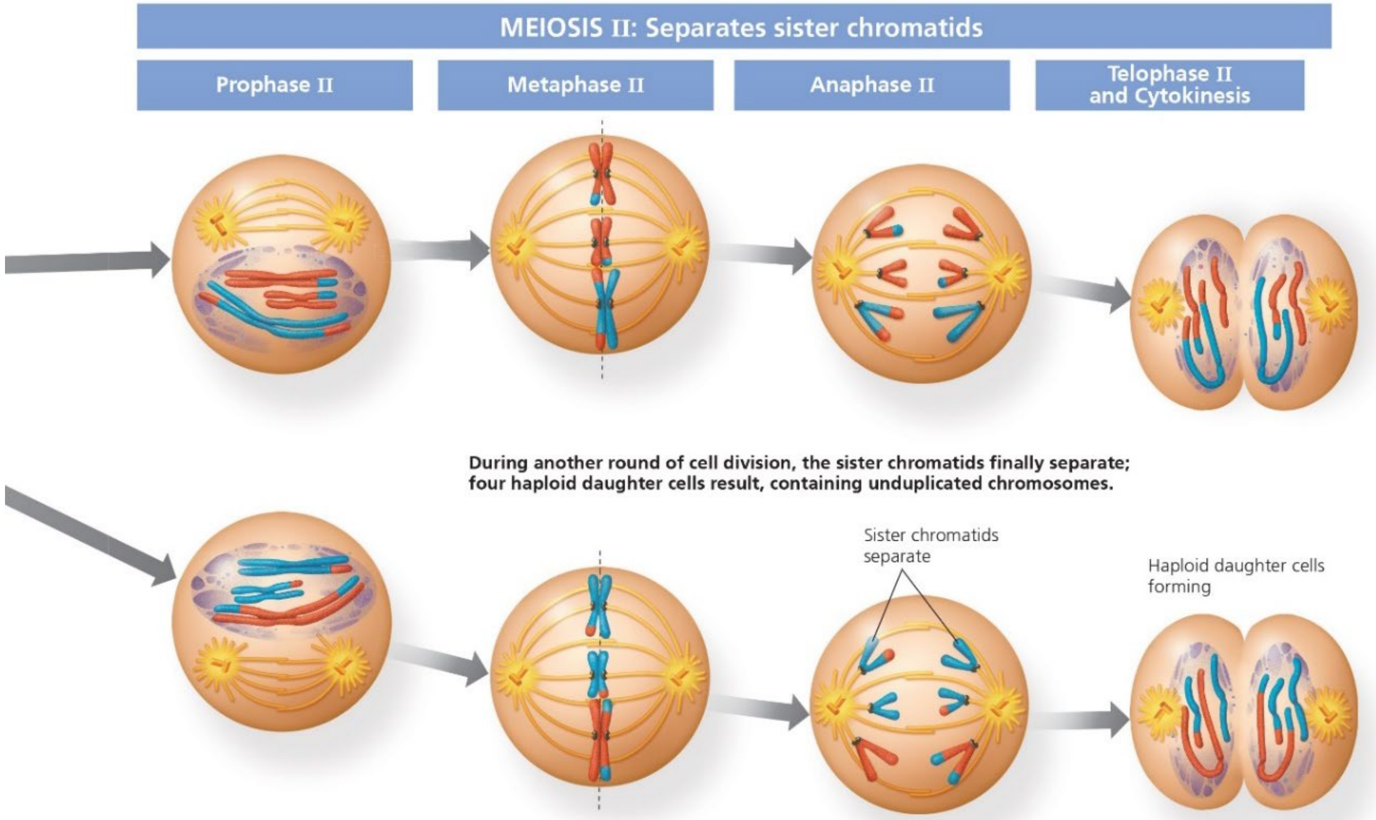

Prophase II

- A spindle apparatus forms.
- In late prophase II (not shown here), chromosomes, each still composed of two chromatids associated at the centromere, are moved by microtubules towards the metaphase II plate.

Metaphase II

- The chromosomes are positioned at the metaphase plate as in mitosis.
- Because of crossing over in meiosis I, the two sister chromatids of each chromosome are not genetically identical.
- The kinetochores of sister chromatids are attached to microtubules extending from opposite poles.

Anaphase II

- Breakdown of proteins holding the sister chromatids together at the centromere allows the chromatids to separate. The chromatids move towards opposite poles as individual chromosomes.

Telophase II and Cytokinesis

- Nuclei form, the chromosomes begin decondensing, and cytokinesis occurs.
- The meiotic division of one parent cell produces four daughter cells, each with a haploid set of (unduplicated) chromosomes.
- The four daughter cells are genetically distinct from one another and from the parent cell.

MAKE CONNECTIONS ➤ *Look at Figure 12.7 and imagine the two daughter cells undergoing another round of mitosis, yielding four cells. Compare the number of chromosomes in each of those four cells, after mitosis, with the number in each cell in Figure 13.8, after meiosis. What is it about the process of meiosis that accounts for this difference, even though meiosis also includes two cell divisions?*

BioFlix® Animation: Meiosis

From Urry LA et al. Campbell Biology, Australian and New Zealand edition. 11th edn. Sydney, Australia: Pearson Australia, 2017, p263

Meiosis vs mitosis

What are the key differences between the two processes?

- Meiosis reduces the number of chromosome sets from two (diploid) to one (haploid), whereas mitosis conserves the number of chromosome sets.
- Meiosis produces cells that differ genetically from their parent cell and from each other, whereas mitosis produces daughter cells that are genetically identical to their parent cell and to each other.
- Three events unique to meiosis occur during meiosis I:

  1. Synapsis and crossing over
  2. Alignment of homologous pairs at the metaphase plate
  3. Separation of homologues

Summary

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- An appreciation of the basic structures and functions of eukaryotic (and prokaryotic) cells can inform many aspects of dentistry, e.g. antibiotic usage
- Cells are the basic structural and functional units of every organism
- They are of two distinct types: prokaryotic and eukaryotic
- The nucleus contains most of the genes in the eukaryotic cells.
- Prokaryotes undergo cell division by binary fission
- Eukaryotes undergo cell division by the cell cycle
- The cell cycle consists of four phases – G1, S, G2 and M
- Asexual reproduction occurs by mitosis
- In sexual reproduction, two haploid gametes fuse to form a zygote
- Haploid gametes are produced by the process of meiosis

Questions?

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Thank you

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Slides adapted from
Drs Bartle, Hosseinpour and Fernandez (2018-2025)